

# Lecture 26: The "Ford F-150" of Nuclear Energy (The Standard Commercial PWR)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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## Introduction: Standardization

While early test reactors explored exotic geometries, coolants, and breeder configurations, the commercial utility industry rapidly coalesced around a simpler, highly robust design.

- **The Winner:** The Pressurized Water Reactor (PWR), originally designed by Admiral Rickover for naval propulsion, was scaled up for civilian power by Westinghouse (and later Framatome/Areva).
- **Dominance:** Today, roughly 65% of the world's commercial power reactors are PWRs.
- **Key Philosophy:** Keep the water *liquid*. Do not allow bulk boiling in the core. Strictly separate the radioactive primary loop from the clean secondary steam cycle.

## 1 The Scale of the Machine

To put the neutronic and thermal math into perspective, we must first understand the physical scale of a standard commercial PWR (often referred to as the 1000 MWe class).

Despite generating enough energy to power a million homes, the actual nuclear core is surprisingly compact, which explains why our thermal constraints are so severe.

- **Thermal Power:**  $\approx 3000$  MWth (Total heat generated by fission in the core. This corresponds to volumetric power densities on the order of  $10^8$  W/m<sup>3</sup>, comparable to rocket engines.)
- **Electric Power:**  $\approx 1000$  MWe ( $\approx 33\%$  efficiency due to losses and the thermodynamic Carnot limit of the secondary steam cycle)
- **Core Diameter:**  $\approx 3.5$  meters (11.5 feet)
- **Active Core Height:**  $\approx 4.0$  meters (12 to 14 feet)
- **Fuel Assemblies:**  $\approx 157$  to 193 (Depending on a 3-loop or 4-loop Westinghouse design)

## 2 The Primary Loop Architecture

The PWR is defined by its "Two-Loop" system. The highly radioactive water remains permanently sealed inside the thick concrete containment building.

## 2.1 The Primary Circuit

Water at high pressure ( $\approx 15.5$  MPa / 2250 psia) circulates through the core, absorbing massive amounts of heat but remaining subcooled (no bulk boiling).

- $T_{in}$  (Cold Leg)  $\approx 290^\circ\text{C}$
- $T_{out}$  (Hot Leg)  $\approx 325^\circ\text{C}$

## 2.2 The "Lung" of the System: The Pressurizer

Because liquids are functionally incompressible, a solid water loop would experience catastrophic pressure spikes from even tiny temperature fluctuations. To absorb these surges, the PWR relies on the **Pressurizer**.

- **Location:** Connected via a surge line to one of the Hot Legs.
- **State:** It is the *only* place in the primary system where water deliberately exists at saturation ( $T_{sat} \approx 345^\circ\text{C}$ ). It contains a steam bubble at the top (cushion) and liquid at the bottom. Electric heaters boil water to increase pressure (both at startup and during operation), while cold-water sprayers connected to the cold leg return line (specifically the pump discharge where pressure is maximum in the entire system) condense steam to lower pressure.

### Safety Warning: The "Solid" Plant

If the steam bubble in the pressurizer collapses (the plant goes "solid"), the primary loop loses its compressibility. Because water is incompressible, minor temperature fluctuations can cause catastrophic pressure spikes. At Three Mile Island (1979), operators prematurely throttled safety injection because they feared "going solid," unaware that the high level in the pressurizer was actually being caused by steam voids in the core pushing water upward.

## 2.3 The Steam Generators (The Massive Heat Exchangers)

The hot primary water flows through thousands of inverted U-tubes inside the Steam Generator (SG), acting as the critical boundary between the radioactive core and the conventional turbine plant.

- **Primary Side (Tube Side):** Hot, radioactive water from the reactor flows inside the tubes at roughly 2250 psia.
- **Secondary Side (Shell Side):** Clean feedwater surrounds the outside of the tubes. Operating at a lower pressure ( $\approx 1000$  psia), this secondary water boils on the tube surfaces, generating clean saturated steam that is sent out to the turbines.
- **The Pressure Boundary Challenge:** To maximize heat transfer, the walls of the 3,000 to 10,000 tubes are incredibly thin (about 0.04 to 0.05 inches). This paper-thin metal is the *only* physical barrier preventing a high-pressure radioactive leak into the secondary steam system (an event known as a Steam Generator Tube Rupture).
- **The "Denting" Crisis:** In early legacy PWRs, the tubes were held in place by carbon steel support plates with tightly drilled circular holes. As the secondary water boiled, chemical impurities concentrated in the tiny crevices between the tube and the plate. The carbon

steel corroded into magnetite (rust), which expands as it forms. This expanding rust literally crushed and "dented" the tubes inward until they cracked, forcing multi-billion-dollar component replacements across the industry.

- **The Modern Engineering Fix:** Replacement Steam Generators (RSGs) solved this using a mix of materials science and fluid dynamics. Tubes were upgraded to highly corrosion-resistant Inconel 690. Furthermore, instead of tight circular holes, modern stainless steel support plates use "broached" (cloverleaf-shaped) holes. The tube touches the plate at only a few points, leaving open channels that allow high-velocity water to constantly sweep the surface, preventing corrosive chemicals from ever settling.

## 2.4 The Reactor Coolant Pumps (RCPs)

To move thousands of gallons per second, PWRs use massive, vertical, single-stage centrifugal pumps.

- **The Basics and the Seal Problem:** Think of a standard pool pump: an electric motor drives a shaft connected to an impeller. As the impeller rapidly rotates, fluid is sucked in axially and thrown out radially due to centrifugal forces. In this standard design, the motor is "dry" while the impeller is "wet." This means the rotating shaft must penetrate the pump casing, requiring a dynamic rotating seal. In a commercial PWR, that seal must withstand 2250 psia of pressure. This is a major vulnerability, as high-pressure seals inherently wear and leak—which in this case means leaking radioactive primary coolant.
- **The Early Solution (Canned Motor Pumps):** To avoid this leakage entirely, the earliest reactors (like the Shippingport commercial prototype) and all U.S. Navy nuclear vessels utilized "seal-less" or canned motor pumps. In this design, the rotating shaft, impeller, and motor rotor are completely submerged inside the high-pressure primary water. A thin, non-magnetic alloy cylinder (the "can") separates the wet rotor from the dry electrical stator on the outside. The magnetic field passes right through the metal can to spin the rotor, meaning no physical shaft ever penetrates the pressure boundary.
- **The Coastdown Challenge and Legacy RCPs:** As commercial PWRs scaled up to 1,000+ MWe, a physics problem emerged: the *coastdown requirement*. If a plant loses power, the pumps cannot stop instantly; they must slowly spin down over several minutes to keep coolant flowing while the control rods insert.
  - *The Flywheel Problem:* To achieve this momentum, engineers needed to add a heavy steel flywheel to the pump shaft. However, placing a giant steel flywheel *inside* a water-filled canned motor creates immense fluid drag, destroying the pump's efficiency. (*Note: You cannot just make the impeller itself massive, because thickening the blades or increasing its diameter would ruin the pump's hydraulic efficiency and choke the water flow!*)
  - *The Shaft-Seal Compromise:* To keep the giant flywheel spinning in dry air where friction is low, the industry moved back to traditional motors outside the pipe. Developing the complex, multi-stage, controlled-leakage shaft seals to prevent a Loss of Coolant Accident (LOCA) along this shaft became one of the greatest mechanical engineering achievements (and vulnerabilities) of the legacy PWR.
- **The Modern Solution (AP1000):** Generation III+ designs (specifically the Westinghouse AP1000) eliminate the shaft leakage issue entirely by returning to the canned motor pump design, but with a modern materials science twist.

- *Solving the Friction Issue:* To achieve the required coastdown momentum without creating massive fluid drag, the AP1000 pumps abandon traditional steel flywheels. Instead, they utilize internal flywheels encased in a smooth shell and made of ultra-dense **Tungsten heavy alloy** (and historically, depleted uranium). This provides massive rotational inertia in a highly compact, small-diameter profile, allowing it to spin inside the water with minimal hydraulic resistance.

### 3 Reactor Vessel and Containment: The Defense in Depth

The PWR relies on multiple physical barriers to prevent the release of fission products. This "Defense in Depth" philosophy is personified by the two most massive components of the plant. Understanding the extreme difference in scale between these two structures is the key to understanding nuclear thermodynamics.

#### 3.1 The First Barrier: The Reactor Pressure Vessel (RPV)

The RPV is the heart of the primary system. It is essentially a massive, vertically oriented steel thermos holding the core.

- **Dimensions:** It is relatively compact, roughly 4.5 meters (15 feet) in diameter and 12 to 14 meters (40 to 45 feet) tall.
- **The Armor:** To withstand 15.5 MPa (2250 psi) at 325°C without bursting, the walls in the "beltline" region (directly surrounding the core) are forged from low-alloy carbon steel (e.g., SA-508) that is **20 to 25 cm (8 to 10 inches) thick**.
- **Cladding:** The inner surface is "battered" with a thin layer of stainless steel to prevent the corrosive, high-temperature primary water from rusting the carbon steel.
- **Neutron Embrittlement:** Over decades of operation, fast neutrons from the core displace atoms in the steel lattice, making the metal brittle. The embrittlement rate depends strongly on trace alloying elements (e.g., copper and nickel content). This shift in the metal's "Nil-Ductility Transition Temperature" makes the vessel more susceptible to pressurized thermal shock (cracking if flooded with cold water while under pressure). Because the RPV is the only component in the plant that essentially cannot be replaced, embrittlement is the primary limiting factor for extending plant life beyond 60 years. Utilities monitor this by placing "surveillance capsules" (small samples of the same steel) inside the vessel and periodically pulling them out for destructive testing.

#### 3.2 The "Brute Force" Barrier: Large Dry Containment

If the primary loop ruptures (a Large Break LOCA), 2250 psi water instantly flashes to steam, expanding its volume by a factor of order 1000. The containment building must catch this thermodynamic blowdown.

- **The Concept:** Containment by sheer volume. It is a massive cavern, often 130 feet in diameter and 200 feet tall (mostly empty air), acting as a giant expansion tank.
- **Design Limit:** Engineered to withstand a peak internal pressure of roughly **60 psig** during a LOCA.

- **Construction:** 3.5 to 4 feet of heavily reinforced (and often post-tensioned) concrete. The interior is lined with a 1/4-inch carbon steel plate acting as a leak-proof gas barrier, not a structural support.
- **Local Example:** The **Palisades Nuclear Plant** utilizes a classic Large Dry containment. TMI-2 also used this design, which is why its massive internal volume was able to easily absorb the 28 psig pressure spike of a severe hydrogen deflagration in 1979 without failing.

### 3.3 The "Vacuum" Modification: Sub-Atmospheric Containment

To reduce the cost of large dry containment domes, some plants (designed primarily by the engineering firm Stone & Webster) utilize a thermodynamic trick: altering the initial starting pressure to reduce the peak accident pressure.

- **The Concept:** The concrete dome is physically similar to a Large Dry containment, but during normal operation, pumps draw the internal atmosphere down to a partial vacuum (typically **9 to 11 psia**).
- **The Thermodynamic Benefit:** If a LOCA occurs, the flashing steam adds a massive, fixed amount of pressure. Because the baseline pressure is artificially lowered, the peak accident pressure only reaches roughly **40 to 45 psig** (compared to the standard 60 psig). This allows for a slight reduction in concrete thickness and structural costs.
- **The Safety Benefit (Zero Out-Leakage):** Fluids naturally flow from high to low pressure. During normal operation, any tiny leak in the containment pulls clean outside air *inward*, completely preventing radioactive gas from escaping. Following a LOCA, containment spray systems rain down cold water to condense the steam. This rapidly collapses the pressure, snapping the building back into a sub-atmospheric vacuum within 1 to 4 hours, securely trapping all fission products inside.
- **The Trade-off:** The major operational disadvantage is accessibility. Maintaining a massive concrete dome at a constant vacuum requires heavy-duty continuous pumping. Furthermore, the sub-atmospheric environment means workers cannot easily or safely enter the containment building to perform maintenance while the reactor is running. These disadvantages resulted in the approach not being widely used today due to operational complexity.
- **Example:** The Surry and North Anna Power Stations in Virginia.

### 3.4 The "Thermodynamic Trick": Ice Condenser Containment

Because 4-foot-thick prestressed concrete is incredibly expensive, Westinghouse engineers developed a thermodynamic workaround to further shrink the containment building.

- **The Concept:** Instead of an empty dome, the building is divided into an upper and lower compartment, separated by a cold-storage ring holding **2 to 3 million pounds of borated ice**.
- **The Mechanism:** If a pipe breaks, the expanding steam forces open spring-loaded doors and blasts through the ice. The ice instantly melts, absorbing the latent heat of vaporization and quenching the steam.

- **The Trade-off:** Because the steam is instantly condensed, the peak accident pressure drops to only **12 to 15 psig**. This allows for a physically smaller building with a much thinner steel shell and less concrete.
- **The Safety Catch:** A smaller volume means any hydrogen generated during a core melt concentrates much faster into an explosive mixture. Therefore, Ice Condensers are legally required to maintain active **hydrogen igniters or recombiners** to safely burn off hydrogen in small, controlled puffs.
- **Local Example:** The **Donald C. Cook Nuclear Plant** in Bridgman, MI utilizes the Ice Condenser design for both of its units.

## 4 Fuel Management and Burnup

Commercial utilities do not replace all the fuel at once. Instead, they treat the core like a rotating inventory.

### 4.1 The 18-Month Cycle

Every 18 to 24 months, the plant shuts down for refueling.

- **The 1/3 Core Rule:** Typically, only the "oldest" third of the fuel assemblies (those that have been in for 3 cycles) are removed and sent to the spent fuel pool.
- **Shuffling:** The remaining 2/3 are shuffled to new positions (often moving from the center to the periphery) to ensure an even power distribution.

### 4.2 Defining "Burnup"

We measure how much energy we extract from the fuel in units of **Megawatt-days per Metric Ton of Uranium (MWd/MTU)**.

- **Modern PWR Limit:**  $\approx 45,000$  to  $60,000$  MWd/MTU.
- **Efficiency:** High burnup is desirable for economics, but is limited by "cladding creep" and the buildup of gaseous fission products (Xe, Kr) inside the fuel rod, which increases internal pressure.

## 5 Typical Plant Power Balance

To understand the "Engineering Efficiency," we look at where the energy goes. A standard 4-loop Westinghouse plant might look like this:

- **Net Thermal Efficiency:**  $\eta = \frac{W_{net}}{Q_{th}} \approx 32.5\%$ .
- **Heat Sink:** The remaining  $\approx 2300$  MW of heat is rejected to the environment (river, lake, or cooling tower).

| Component                                         | Power Level      | Notes                             |
|---------------------------------------------------|------------------|-----------------------------------|
| Core Thermal Power ( $Q_{th}$ )                   | 3411 MWth        | Total heat from fission/decay     |
| RCP Heat Input                                    | +14 MWth         | Large pumps actually add heat!    |
| <b>Total Steam Heat</b>                           | <b>3425 MWth</b> | Delivered to Steam Generators     |
| Gross Electric Output ( $W_{gross}$ )             | 1150 MWe         | Generated by the Main Turbine     |
| House Loads (Pumps/Fans)                          | -40 MWe          | Electricity used to run the plant |
| <b>Net Electric Output (<math>W_{net}</math>)</b> | <b>1110 MWe</b>  | What actually goes to the grid    |

## 6 Core Geometry: Driven by Thermal Limits

Before we look at neutronics, we must build the core grid. As we derived in Lecture 25, the size of the fuel rods is strictly dictated by heat transfer, not nuclear physics.

### 6.1 Recall: The $R^2$ Limit

Uranium Dioxide ( $UO_2$ ) is the standard fuel because it is chemically stable and resists radiation swelling. However, it is a ceramic with very poor thermal conductivity ( $k_f \approx 3.0 \text{ W/m} \cdot \text{K}$ ).

Recall the centerline temperature derivation for a solid cylinder with internal heat generation ( $q'''$ ):

$$\Delta T_{\text{pellet}} = T_{\text{center}} - T_{\text{surface}} = \frac{q''' R^2}{4k_f} \quad (1)$$

Because the internal temperature drop scales with the **square of the radius** ( $R^2$ ), we cannot simply build "fat" fuel rods to hold more uranium. If we doubled the radius of a standard rod, the internal  $\Delta T$  would quadruple, causing the center of the  $UO_2$  pellet to melt ( $\approx 2800^\circ\text{C}$ ). Therefore, thermal conduction strictly limits the fuel to a "thin rod" geometry (roughly 1 cm in diameter).

## 7 The Core Physics: Mean Free Path and Moderation

Once the rod size is fixed by thermal limits ( $\sim 1 \text{ cm}$ ), we tune the *spacing* (pitch) between the rods based on neutronics.

### 7.1 The Fast Neutron (2 MeV): Transparent Lattice

When a neutron is born from fission, it has high energy.

- **Physics:** The scattering Mean Free Path ( $\lambda_s$ ) in water is  $\approx 3$  to **5** cm.
- **Geometry:** The Fuel Rod Pitch (center-to-center distance) is only **1.26** cm.
- **Result:** The fast neutron essentially "ignores" the physical lattice structure, traveling past 4 or 5 rods before undergoing significant collision.

### 7.2 The Thermal Neutron (0.025 eV): Optimizing the Gap

Once the neutron becomes thermal (slowed down), its behavior changes drastically.

- **Physics:** The Mean Free Path in water drops significantly to  $\lambda_{\text{thermal}} \approx 0.3$  cm.

- **Geometry:** The actual physical water gap between the rods is roughly **0.31** cm.
- **Optimization (The Under-Moderated Core):** This physical gap is strictly designed so that the Hydrogen-to-Uranium (H/U) ratio ensures the core is slightly **under-moderated**. This means there is slightly less water than is optimal for perfect neutron slowing. This is a critical safety feature: if the water heats up and expands (decreasing its density), the moderation gets even worse, and the reactor power naturally drops (required negative temperature coefficient for reactivity).

## 8 The Assembly Structure: The 17x17 Standard

Commercial reactors don't load individual pins; they load massive, highly engineered bundles. The industry standard (Westinghouse) uses a  $17 \times 17$  square lattice.

- **Dimensions:**
  - Rod Diameter: 0.950 cm.
  - Clad Thickness: 0.057 cm (Zircaloy-4 or ZIRLO).
  - Array: 264 Fuel Rods, 24 Guide Thimbles, 1 Central Instrument Tube.
- **Guide Thimbles:** These are empty tubes replacing fuel rods in specific locations. They provide a clear structural channel for control rods or instruments to be inserted directly into the assembly.
- **Spacer Grids & Mixing Vanes:** To keep the 12-foot-long rods from bowing and touching each other, they are held by spring-loaded Zircaloy/Inconel grid straps. Furthermore, these grids contain small **mixing vanes** that induce swirl in the water, stripping away the boiling boundary layer to prevent DNB (as discussed in Lecture 25).

## 9 Control Strategy 1: The Mechanical "Spider" (RCCA)

Early reactors used large cruciform blades that caused severe "flux dipping" and localized power spikes. The modern PWR solves this with the **Rod Cluster Control Assembly (RCCA)**.

- **Design:** 24 thin absorber rods (made of Silver-Indium-Cadmium) are attached to a common top hub (the "spider"). These fingers slide perfectly down into the 24 Guide Thimbles of the fuel assembly.
- **Advantage:** Distributing the absorber material evenly throughout the assembly keeps the neutronic power profile much flatter.
- **Safety Mechanism:** They are held vertically above the core by electromagnetic coils. In the event of a power failure, the magnets release, and gravity drops the rods into the core (a fail-safe SCRAM).

## 10 Control Strategy 2: Chemical Shim and Burnable Poisons

Mechanical rods warp the axial flux profile. Therefore, in a commercial PWR, mechanical rods are mostly withdrawn during normal operation. They are used for startup, shutdown, and safety transients—**not** for slow burnup compensation over the 18-month cycle.

## 10.1 Soluble Boron (The ChemE Approach)

To hold down the massive excess reactivity of a fresh core evenly, we dissolve **Boric Acid** ( $H_3BO_3$ ) directly into the primary coolant. As the U-235 burns up over 18 months, the operators slowly dilute the boron concentration to maintain criticality.

- **Boron Letdown Curve:** Starts at  $\approx 1200$  ppm (Start of Cycle) and is diluted to  $\approx 10$  ppm (End of Cycle).

## 10.2 The Safety Constraint: Moderator Temperature Coefficient

Using chemical shim (boron) introduces a severe, counter-intuitive physics constraint:

- **The Normal Safety Mechanism:** Normally, if reactor water heats up and expands, its density drops. Less dense water means less moderation, which safely lowers reactor power (a negative Moderator Temperature Coefficient).
- **The Boron Problem:** However, if the water is heavily poisoned with Boric Acid, that expanding water also pushes Boron *out* of the core. Removing a neutron poison *increases* reactivity!
- **The Limit:** If the boron concentration is too high, the positive reactivity from removing the poison will overpower the negative reactivity from losing the moderator. The core would accelerate as it heats up! Therefore, the NRC strictly limits the maximum allowable soluble boron concentration to ensure the net Moderator Temperature Coefficient always remains negative under operating conditions.

## 10.3 The Solution: Burnable Poisons

If we want to run a reactor for 24 months (requiring higher initial enrichment), the soluble boron concentration needed to control excess reactivity would challenge safety constraints on the moderator temperature coefficient. To address this, engineers incorporate neutron-absorbing materials directly into the fuel.

- These are called **Integral Burnable Absorbers (IBAs)**.
- In one approach, a thin coating of boron-10 (typically  $ZrB_2$ ) is applied to the surface of selected fuel pellets—these are specifically called **Integral Fuel Burnable Absorbers (IFBAs)**.
- Alternatively, neutron absorbers such as gadolinium or erbium are mixed directly into the  $UO_2$  fuel matrix.

These materials suppress excess reactivity at the beginning of the fuel cycle, allowing operation with lower soluble boron concentrations. As the cycle progresses, the absorbers are gradually depleted, offsetting fuel burnup and helping maintain a controlled reactivity profile.

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